Modular Contradiction and Diophantine Equations

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1 Introduction

Take a look at this Diophantine equation:

$$x^3 + 117y^3 = 5.$$

During the 1970s, mathematicians were stumped for years in finding integer solutions for this equation. One mathematician proved that there can only be at most 18 possible solutions. Later, another mathematician used algebraic number theory to finally show that it has no solutions at all. But we can also solve this equation simply by using Modular Arithmetic.

Observe that if we reduce the equation (mod 9), then we get:

$$x^3 \equiv 5 \pmod{9}.$$

However, we know that the residues (mod 9) for x^3 can only be $0, \pm 1$. Thus, this easily shows that there does not exist any integer solutions for this equation.

We can see how powerful checking the modulo in solving Diophantine equations. This kind of method is what we call **Modular Contradiction**.

Modular arithmetic is very helpful for studying equations especially when dealing with powers. For example, we know that for perfect squares $x^2 \equiv \{0, 1\} \pmod{3}$. These kinds of relations are very simple facts but are often useful in helping us reduce the equations and finding solutions. While these kind of methods don't always give a complete answer, they frequently narrow down possible solutions significantly.

Modular Contradictions are often useful when solving equations with finite or no solutions. The key idea is to pick a modulus m such that the equation leads to an impossible congruence (e.g. the example above). Once we choose the appropriate modulus, checking that there are no integer solutions is often very straight-forward or we will be able to deduce some necessary conditions the solutions satisfy. Thus, the main difficulty in solving these types of problems is deciding which modulus to use.

Here are some suggestions in choosing an appropriate modulus:

1. *Choose a modulus that vanishes some coefficients.* This often simplifies the equation and can help to eliminate variables.
2. *Choose a modulus with very few n^{th} power residues.* Below are some helpful relations for powers that are often used:

- $x^2 \equiv \{0, 1\} \pmod{3}$
- $x^2 \equiv \{0, \pm 1\} \pmod{5}$
- $x^2 \equiv \{0, 1, 4\} \pmod{8}$
- $x^3 \equiv \{0, \pm 1\} \pmod{9}$
- $x^3 \equiv \{0, \pm 1\} \pmod{7}$
- $x^4 \equiv \{0, 1\} \pmod{16}$
- $x^4 \equiv \{0, 1, 5, 16\} \pmod{20}$
- $x^5 \equiv \{0, \pm 1\} \pmod{11}$

In general, these are results from well-known number theory theorems. It is essential to know these theorems and corollaries by heart in solving number theory problems.

Fermat's Little Theorem. Let p be a prime number, and x be any integer. Then $x^p - x$ is always divisible by p or

$$x^p \equiv x \pmod{p}.$$

Furthermore, if x is not divisible by p, then

$$x^{p-1} \equiv 1 \pmod{p} \Rightarrow x^{\frac{p-1}{2}} \equiv \pm 1 \pmod{p}.$$

Corollary. If p is a Sophie Germain prime (p and $2p+1$ is also a prime) and x is a positive integer, then

$$x^p \equiv \{0, \pm 1\} \pmod{2p+1}$$

Euler's Theorem. Let n be a positive integer, and let a be an integer that is relatively prime to n, then

$$a^{\phi}(n) \equiv 1 \pmod{n},$$

where $\phi(n)$ is Euler's totient function that counts the number of positive integers less than n that are coprime to n.

Note that Fermat's Little Theorem is a special case of Euler's Theorem when n=p is prime.

Wilson's Theorem. Let p be prime number, then

$$(p-1)! \equiv -1 \pmod{p}$$

When a problem says to prove a diophantine equation has no solutions it usually has a modular solution. Finding the right mod is quite an important trick. Find the modulo which reduces the cases to as small as possible and this will likely be the right mod to work with. If not, try many different mods and see which ones work.

When dealing with factorials, it is often advantageous to take advantage of the fact that $m! \equiv 0$ (mod p) for all positive integers $m \geq p$. By finding the right mod, we can reduce the number of cases significantly.

2 Problems

Problem 1. Find all pairs of integers (x, y) that satisfy the-equation

$$x! = y^4 + 2025.$$

Problem 2. Find all positive integers $x, y, z > 1$ such that

$$1! + 2! + \ldots + x! = y^z.$$

Problem 3. (USAJMO 2013) Are there integers a, b such that $a^5 b + 3$ and $ab^5 + 3$ are perfect cubes?

Problem 4. Prove that the following numbers cannot be expressed as the sum of the cubes of some consecutive integers.

(1) 385^{97} (2) 366^{17}

Problem 5. (Hong Kong TST 2002) Prove that if a, b, c, d are integers such that

$$(3a + 5b)(7b + 11c)(13c + 17d)(19d + 23a) = 2001^{2001}$$

then a is even.

Problem 6. Find all powers of 2, such that after deleting its first digit (in decimal system representation) the new number is also a power of 2.

Problem 7. Prove that all positive integer solutions of the equation

$$8^x + 15^y = 17^z$$

are $x = y = z = 2$.

Problem 8. (2024 Mongolia MO) Find all triples (a, b, c) of positive integers such that $a \leq b$ and $a! + b! = c^4 + 2024$.

Problem 9. (2023 JBMO) Find all pairs (a, b) of positive integers such that $a! + b$ and $b! + a$ are both powers of 5.

Problem 10. (2012 IMO Shortlist) Find all triples (x, y, z) of positive integers such that $x \leq y \leq z$ and

$$x^3(y^3 + z^3) = 2012(xyz + 2)$$

Problem 11. (2019 IMO) Find all pairs (k, n) of positive integers such that

$$k! = (2n - 1)(2n - 2)(2n - 4) \cdots (2n - 2^{n-1}).$$

Problem 12. (2022 IMO) Find all triples (a, b, p) of positive integers with p prime and

$$a^p = b! + p.$$